Perhaps we can do this: if, ten years ago, your sister had left for Proxima b, taking with her a calendar to keep track of the passage of time, can we think that *now* for her is when she has recorded that ten years have passed? No, this does not work either: she might have returned here ten of *her* years after leaving, arriving back where, in the meantime, twenty years have elapsed. So when the hell *is* "now" on Proxima b?

The truth of the matter is that we need to give up asking the question.[27]

There is no special moment on Proxima b that corresponds to what constitutes the present here and now.

Dear reader, pause for a moment to let this conclusion sink in. In my opinion, it is the most astounding conclusion arrived at in the whole of contemporary physics.

It simply makes no sense to ask which moment in the life of your sister on Proxima b corresponds to *now*. It is like asking which football team has won a basketball championship, how much money a swallow has earned, or how much a musical note weighs. They are nonsensical questions because football teams play football, not basketball; swallows do not busy themselves earning money; sounds cannot be weighed. "Basketball champions" refers to a team of basketball players, not to footballers. Monetary profit refers to human society, not to swallows. The notion of "the present" refers to things that are close to us, not to anything that is far away.

Our "present" does not extend throughout the universe. It is like a bubble around us.

How far does this bubble extend? It depends on the precision with which we determine time. If by nanoseconds, the present is defined only over a few meters; if by milliseconds, it is defined over thousands of kilometers. As humans, we distinguish tenths of a second only with great difficulty; we can easily consider our entire planet to be like a single bubble where we can speak of the present as if it were an instant shared by us all. This is as far as we can go.

There is our past: all the events that happened before what we can witness now. There is our future: the events that will happen after the moment from which we can see the here and now. Between this past and this future there is an interval that is neither past nor future and still has a duration: fifteen minutes on Mars; eight years on Proxima b; millions of years in the Andromeda galaxy. It is the expanded present.[28] It is perhaps the greatest and strangest of Einstein's discoveries.

The idea that a well-defined *now* exists throughout the universe is an illusion, an illegitimate extrapolation of our own experience.[29]